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Machine learning and chemometrics-based detection of sugar adulteration in black tea using UV-visible spectroscopy

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Abstract:
Tea (*Camellia sinensis*) is one of the most widely consumed beverages worldwide and ensuring quality is essential for consumer safety and quality control. However, adulteration with low-cost substances such as sugar remains a significant issue in the tea industry. This study presents a rapid, non-destructive approach for detecting sugar adulteration in black tea using UV-Visible spectroscopy combined with chemometrics and machine learning techniques. A total of 250 orthodox black tea samples were collected from seven major tea-growing regions of Sri Lanka, from which 100 representative samples were selected for analysis. Tea and sugar samples were separately ground and sieved, and sugar-adulterated tea samples were prepared gravimetrically at 0.05%–40% (w/w) by homogenization. Spectra were recorded over 190–1100 nm, with five independent replicates. Spectral data were pre-processed using Savitzky-Golay smoothing followed by mean-centering and auto-scaling to reduce noise and improve signal quality. Principal component analysis was applied for exploratory data analysis and dimensionality reduction, revealing that the first two principal components accounted for 79.8% of the total variance. Partial least squares discriminant analysis (PLS-DA) was used for supervised classification, achieving strong model performance with R^2 = 0.866 and Q^2 = 0.851, demonstrating good predictive ability and model robustness. In addition, a support vector machine classifier with a radial basis function kernel was developed, achieving an overall accuracy of 88% and effectively classifying adulterated samples. The PLS-DA model was internally validated using cross-validation, and machine learning models were evaluated using a 70:30 train–test split. The results confirm that integrating UV-Visible spectroscopy with multivariate analysis provides a reliable, rapid, and non-destructive method for detecting sugar adulteration in black tea, with strong potential for application in food quality control and product authentication.

Keywords: Black tea; Chemometrics; Sugar adulteration; Support vector machine; UV-Visible spectroscopy

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